

Exercise 1

My completed code is located inside **rigidbody2d_friction.py**.

Exercise 2

2a)

The slight rocking/sliding is a numerical artifact of the contact solver. We treat each contact independently and then average, rather than solving all contacts simultaneously, so the normal/tangential impulses do not cancel perfectly. With gravity on, each frame injects a small downward velocity, and the discrete integration plus position correction introduces tiny residual errors. Those errors appear as jitter and slow drift even when the theoretical static-friction condition is satisfied.

2b)

For an incline of 13° (0.227 rad), the theoretical threshold for no acceleration is $\mu_s = \tan(13^\circ) \sim 0.231$. When μ_s is well above this value, the box should be stationary; in the simulation, it still jitters and may drift at nearly constant speed due to numerical issues in 2a. As μ_s approaches ~ 0.23 , the drift becomes more noticeable. Once μ_s drops below ~ 0.23 , the box should transition to clear acceleration down the plane rather than just a slow constant-speed slide.

2c)

My code for this exercise is written in **rigidbody2d_friction_extra.py**.

I improved stability by adding a simple sleep/resting state and keeping full tangential position correction only when the contact is inside the static friction cone. Concretely, I track small velocities over several frames and, once both linear and angular speeds stay below a small threshold for a few frames, I freeze the body by zeroing vel/omega and disabling gravity for that body. This removes the residual jitter caused by gravity nudging the contact every frame. When the contact is sliding (outside the static cone), I only apply a scaled tangential position correction so the block can still move; when it's static, I fully undo tangential drift. This combination reduces rocking/sliding at the critical angle without breaking normal sliding behaviour.

Exercise 3

3a)

My code for this exercise is written in **rigidbody2d_rolling.py**

Exercise 4

The code written in Exercise 3 has an implementation where, if you press the “N” key, it restarts the race and switches the scenario. We will complete the race without friction. The results are shown below:

```
raceMode=0 (hollow cylinder)
finish speed |v| = 2.433307
raceMode=1 (solid cylinder)
finish speed |v| = 2.821543
raceMode=2 (sphere)
finish speed |v| = 2.911135
raceMode=3 (sliding particle)
finish speed |v| = 3.116057
```

As we can see, the ordering of the race, from slowest to fastest, is:

1. Sliding particle (no rotation) is fastest.
2. Sphere ($I = \frac{2}{5} MR^2$)
3. Solid cylinder ($I = \frac{1}{2} MR^2$)
4. Hollow cylinder ($I = MR^2$) is slowest.

This is the ordering we expect based on the moments of inertia provided in the assignment handout.